

# PROJECT REPORT

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## Customer Shopping Behaviour Analysis

### Using Python, SQL, and Power BI

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#### 1. Introduction

In today's data-driven environment, understanding customer behaviour is essential for improving business performance, marketing strategies, and customer retention.

This project focuses on analysing customer shopping behaviour using a dataset obtained from Kaggle. The analysis is performed using **Python for data preprocessing, SQL for querying and analysis, and Power BI for visualization and dashboard creation**.

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#### 2. Dataset Description

- **Source:** Kaggle
- **Total Records:** 3,900+
- **Total Features:** 18

##### Key Attributes:

- Customer ID
- Age, Gender
- Item Purchased, Category
- Purchase Amount (USD)
- Location
- Season
- Review Rating

- Subscription Status
  - Payment Method
  - Frequency of Purchases
- 

### 3. Data Preprocessing (Python)

#### Tools Used:

- Pandas
- **Steps Performed:**
  - Loaded dataset using Pandas
  - Checked structure using `.info()` and `.describe()`
  - Identified missing values

#### Handling Missing Values:

- Filled missing values in review rating using **median grouped by category**

#### Column Standardization:

- Converted column names to lowercase
- Replaced spaces with underscores

#### Column Renaming:

- Renamed `purchase amount_(usd)` to `purchase_amount`
- 

### Screenshot 1: Raw Dataset

4]: df.head()

4]:

|   | Customer ID | Age | Gender | Item Purchased | Category | Purchase Amount (USD) | Location      | Size | Color     | Season | Review Rating | Subscription Status | Shipping Type | Discount Applied | Promo Code Used | Previous Purchases | Payment Method |
|---|-------------|-----|--------|----------------|----------|-----------------------|---------------|------|-----------|--------|---------------|---------------------|---------------|------------------|-----------------|--------------------|----------------|
| 0 | 1           | 55  | Male   | Blouse         | Clothing | 53                    | Kentucky      | L    | Gray      | Winter | 3.1           | Yes                 | Express       | Yes              | Yes             | 14                 | Ver            |
| 1 | 2           | 19  | Male   | Sweater        | Clothing | 64                    | Maine         | L    | Maroon    | Winter | 3.1           | Yes                 | Express       | Yes              | Yes             | 2                  | C              |
| 2 | 3           | 50  | Male   | Jeans          | Clothing | 73                    | Massachusetts | S    | Maroon    | Spring | 3.1           | Yes                 | Free Shipping | Yes              | Yes             | 23                 | Cr C           |
| 3 | 4           | 21  | Male   | Sandals        | Footwear | 90                    | Rhode Island  | M    | Maroon    | Spring | 3.5           | Yes                 | Next Day Air  | Yes              | Yes             | 49                 | Pa             |
| 4 | 5           | 45  | Male   | Blouse         | Clothing | 49                    | Oregon        | M    | Turquoise | Spring | 2.7           | Yes                 | Free Shipping | Yes              | Yes             | 31                 | Pa             |

Screenshot 2: Dataset Info

[7]: df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                       3900 non-null   object
4   Category                             3900 non-null   object
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   object
7   Size                                 3900 non-null   object
8   Color                                3900 non-null   object
9   Season                               3900 non-null   object
10  Review Rating                        3883 non-null   float64
11  Subscription Status                  3900 non-null   object
12  Shipping Type                       3900 non-null   object
13  Discount Applied                    3900 non-null   object
14  Promo Code Used                     3900 non-null   object
15  Previous Purchases                   3900 non-null   int64
16  Payment Method                      3900 non-null   object
17  Frequency of Purchases               3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

Screenshot 3: Missing Values

```
[9]: df.isnull().sum()

[9]: Customer ID      0
     Age            0
     Gender         0
     Item Purchased  0
     Category        0
     Purchase Amount (USD) 0
     Location        0
     Size           0
     Color          0
     Season         0
     Review Rating   17
     Subscription Status 0
     Shipping Type   0
     Discount Applied 0
     Promo Code Used  0
     Previous Purchases 0
     Payment Method   0
     Frequency of Purchases 0
     dtype: int64

[10]: df['Review Rating']=df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))

[11]: df.isnull().sum()

Customer ID      0
Age            0
Gender         0
Item Purchased  0
Category        0
Purchase Amount (USD) 0
Location        0
Size           0
Color          0
Season         0
Review Rating   0
Subscription Status 0
Shipping Type   0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64
```

## 4. Feature Engineering

New variables were created to enhance analysis:

### 1. Age Group

- Created using `pd.qcut()`
- Categories:
  - Young Adult
  - Adult
  - Middle-aged
  - Senior

```
[15]: # create a age group column
labels = ['Young Adult' , 'Adult' , 'Middle-aged' , 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)

[16]: df[['age', 'age_group']].head()

[16]:
```

|   | age | age_group   |
|---|-----|-------------|
| 0 | 55  | Middle-aged |
| 1 | 19  | Young Adult |
| 2 | 50  | Middle-aged |
| 3 | 21  | Young Adult |
| 4 | 45  | Middle-aged |

## 2. Purchase Frequency (Days)

- Converted categorical frequency into numeric values:
  - Weekly → 7
  - Monthly → 30

```
[17]: # create column purchase_frequency_days
df['frequency_of_purchases'].unique()

[17]: array(['Fortnightly', 'Weekly', 'Annually', 'Quarterly', 'Bi-Weekly',
        'Monthly', 'Every 3 Months'], dtype=object)

[18]: frequenc_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Bi-Weekly': 14,
    'Monthly': 30,
    'Quarterly': 90,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequenc_mapping)
```

## 4. Data Cleaning Enhancement

- Dropped redundant column promo\_code\_used

```
[21]: (df['discount_applied'] == df['promo_code_used']).all()

[21]: np.True_

[23]: df = df.drop('promo_code_used', axis=1)
```

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## Screenshot 4: Feature Engineering Output

```
4]: df[['age_group', 'purchase_frequency_days']].head(10)
```

```
4]:
```

|   | age_group   | purchase_frequency_days |
|---|-------------|-------------------------|
| 0 | Middle-aged | 14                      |
| 1 | Young Adult | 14                      |
| 2 | Middle-aged | 7                       |
| 3 | Young Adult | 7                       |
| 4 | Middle-aged | 365                     |
| 5 | Middle-aged | 7                       |
| 6 | Senior      | 90                      |
| 7 | Young Adult | 7                       |
| 8 | Young Adult | 365                     |
| 9 | Middle-aged | 90                      |

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## 5. Data Loading (MySQL)

- Established connection using SQLAlchemy
- Created database: customer\_behavior
- Loaded cleaned dataset into table: customer

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### Screenshot 5: Data Loaded in MySQL

```
1 • CREATE DATABASE customer_behavior;
2 • USE customer_behavior;
3
4 • SELECT * FROM customer limit 5;
5
```

|   | customer_id | age | gender | item_purchased | category | purchase_amount | location      | size | color     | season | review_rating | subscription_status | shipping_type | discour |
|---|-------------|-----|--------|----------------|----------|-----------------|---------------|------|-----------|--------|---------------|---------------------|---------------|---------|
| ▶ | 1           | 55  | Male   | Blouse         | Clothing | 53              | Kentucky      | L    | Gray      | Winter | 3.1           | Yes                 | Express       | Yes     |
|   | 2           | 19  | Male   | Sweater        | Clothing | 64              | Maine         | L    | Maroon    | Winter | 3.1           | Yes                 | Express       | Yes     |
|   | 3           | 50  | Male   | Jeans          | Clothing | 73              | Massachusetts | S    | Maroon    | Spring | 3.1           | Yes                 | Free Shipping | Yes     |
|   | 4           | 21  | Male   | Sandals        | Footwear | 90              | Rhode Island  | M    | Maroon    | Spring | 3.5           | Yes                 | Next Day Air  | Yes     |
|   | 5           | 45  | Male   | Blouse         | Clothing | 49              | Oregon        | M    | Turquoise | Spring | 2.7           | Yes                 | Free Shipping | Yes     |

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## 6. SQL Analysis

## Concepts Used:

- Aggregation functions (SUM, AVG, COUNT)
  - Subqueries
  - Common Table Expressions (CTE)
  - Window Functions (RANK)
  - CASE statements
  - GROUP BY and filtering
- 

## Key Business Queries:

1. Revenue by Gender
  2. Top 5 highest spending customers
  3. Discount vs spending behavior
  4. Top-rated products
  5. Shipping type comparison
  6. Customer segmentation
  7. Top products per category using window function
  8. Subscription vs spending analysis
- 

## Screenshot 6: Segment customers Query Output

```

80  -- 10.Segment customers into New, Returning, and Loyal based on their
81  -- total number of previous purchases, and show the count of each segment.
82  • with customer_type as (
83      select
84      case
85          when previous_purchases = 1 then "new"
86          when previous_purchases between 2 and 10 then "returning"
87          else "loyal"
88      end as customer_segment
89      from customer
90  )
91  select customer_segment ,count(*) as total_customers
92  from customer_type
93  group by customer_segment;
94

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

|   | customer_segment | total_customers |
|---|------------------|-----------------|
| ► | loyal            | 3116            |
|   | returning        | 701             |
|   | new              | 83              |

## Screenshot 7: Window Function Output

```

94  -- 11. What are the top 3 most purchased products within each category?
95  • select *
96  from (
97      select
98          category,
99          item_purchased,
100         sum(purchase_amount) as total_sales,
101         rank() over(partition by category order by sum(purchase_amount) desc ) as rnk
102      from customer
103      group by category, item_purchased
104  )t
105  where rnk <=3;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

|   | category    | item_purchased | total_sales | rnk |
|---|-------------|----------------|-------------|-----|
| ► | Accessories | Jewelry        | 10010       | 1   |
|   | Accessories | Sunglasses     | 9649        | 2   |
|   | Accessories | Belt           | 9635        | 3   |
|   | Clothing    | Blouse         | 10410       | 1   |
|   | Clothing    | Shirt          | 10332       | 2   |
|   | Clothing    | Dress          | 10320       | 3   |
|   | Footwear    | Shoes          | 9240        | 1   |
|   | Footwear    | Sandals        | 9200        | 2   |
|   | Footwear    | Boots          | 9018        | 3   |
|   | Outerwear   | Coat           | 9275        | 1   |
|   | Outerwear   | Jacket         | 9249        | 2   |

## 7. Power BI Dashboard

### KPIs Created:



- Total Revenue
  - Total Customers
  - Average Purchase Amount
  - Average Review Rating
  - Subscription Rate
  - Revenue per Customer
  - Repeat Customer Rate
  - Revenue with Discount
  - Revenue without Discount
- 

#### **Visualizations Used:**

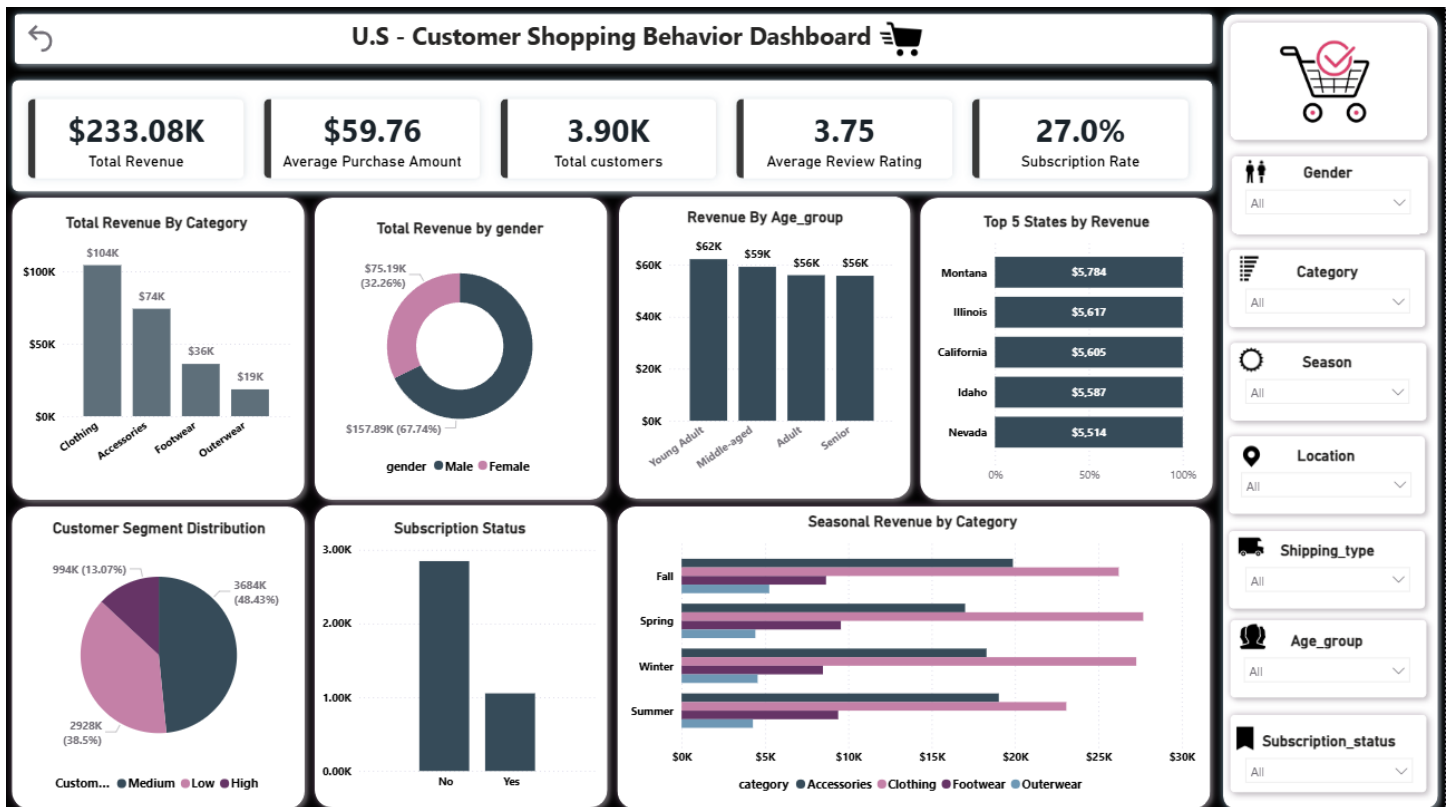
- Bar Charts (Category, Location)
  - Column Charts (Age Group, Subscription)
  - Donut Chart (Gender)
  - Pie Chart (Customer Segment)
  - Clustered Charts (Seasonal Analysis)
- 

#### **Dashboard Features:**

- Interactive slicers:
  - Gender
  - Category
  - Season
  - Location
  - Shipping Type

- Age Group
- Subscription status

## Screenshot 8: Power BI Dashboard



## 8. Key Insights

- Clothing category generates the highest revenue
- Majority of customers are non-subscribers
- Repeat customers contribute significantly to total revenue
- Discounts influence customer purchasing behavior

## 9. Challenges Faced

- Handling missing values in review ratings
- Converting categorical data into numerical format

- Maintaining data consistency
  - Designing an optimized and readable dashboard
- 

## **10. Conclusion**

This project demonstrates a complete data analysis workflow:

- Data cleaning using Python
- Data analysis using SQL
- Visualization using Power BI

The project provides meaningful insights into customer behavior and supports data-driven decision-making.

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## **11. Tools and Technologies**

- Python (Pandas, NumPy)
  - MySQL
  - Power BI
  - SQLAlchemy
- 

## **12. Future Scope**

- Add time-based analysis (time intelligence)
- Build predictive models using machine learning
- Enhance dashboard with real-time data integration